

Schrödinger's Cat in the Holonomic Bifurcation Framework (Neutral Exposition)

This note explains how Schrödinger's cat fits the holonomic bifurcation framework without addressing any specific reader. The goal is to translate the “paradox” into a precise instance of:

- **World A** (theories + branches + restriction),
- a compiler **Gen** (bifurcation $\rightarrow$ LagEvent),
- and scaling via admissible morphisms **Comp** (LagEvent pushforward).

1 World A: Two theories and a restriction map (no quotient assumed)

1.1 Theories and branches

Fix a category of theories $\mathcal{T}\langle$ and a contravariant functor

$$Ext : \mathcal{T}\langle^{op} \rightarrow \mathbf{Set}.$$

For each morphism $i : T \rightarrow T'$ in $\mathcal{T}\langle$, there is a restriction map

$$\rho_i := Ext(i) : Ext(T') \rightarrow Ext(T),$$

interpreted as “forgetting” the additional expressive power of T' down to what T can express. No quotient structure is assumed a priori. The only primitive is the restriction map ρ_i .

1.2 Distinct vs indistinguishable (the key separation)

For $\Delta_0, \Delta_1 \in Ext(T')$, define:

Nontrivial difference

$$Nontriv(\Delta_0, \Delta_1) :\Leftrightarrow \Delta_0 \neq \Delta_1.$$

Agreement under restriction

$$Agree(i) :\Leftrightarrow \rho_i(\Delta_0) = \rho_i(\Delta_1).$$

This separates:

1. “genuinely different micro-branches” from
2. “indistinguishable at the observable/coarse theory level.”

2 Schrödinger’s cat as a canonical relative bifurcation in World A

2.1 Choice of theories

Let:

- T be an outside-the-box observable theory (sealed box, external facts), deliberately not containing any predicate describing the cat’s internal state.
- T' be a refinement/extension of T that adds internal vocabulary, e.g. a predicate $Alive(t_0)$.

Let $i : T \rightarrow T'$ be the corresponding morphism.

2.2 Existence of two coherent branches

Assume the minimal consistency/independence hypothesis:

$$T' + Alive(t_0) \text{ is consistent} \quad \text{and} \quad T' + \neg Alive(t_0) \text{ is consistent.}$$

Then there exist two branches $\Delta_0, \Delta_1 \in Ext(T')$ such that:

$$\Delta_0 \models Alive(t_0), \quad \Delta_1 \models \neg Alive(t_0).$$

2.3 The Schrödinger condition = (Nontriv + Agree)

Nontriv holds: $\Delta_0 \neq \Delta_1$ (they differ on $Alive(t_0)$).

Agree holds along i :

$$\rho_i(\Delta_0) = \rho_i(\Delta_1),$$

because the coarse theory T cannot express $Alive(t_0)$, so the distinction disappears under restriction.

Thus Schrödinger's cat corresponds to an object of the pointed relative-bifurcation data:

$$(i : T \rightarrow T', \Delta_0, \Delta_1, agree, nontriv).$$

3 World B: Primitive holonomy and the obstruction LagEvent

In **PrimitiveHolonomy**, a manipulable bifurcation is captured by an atomic obstruction:

$$LagEvent(\alpha, step) :\Leftrightarrow \exists x \neq x', HolonomyRel(\alpha)(x, x') \wedge Compatible(step, x) \wedge \neg Compatible(step, x').$$

Assume $Compatible(step, x)$ is existential (e.g. $\exists y, sem(step)(x, y)$ or an equivalent transport-witness formulation), so that “being compatible” is witnessed by a concrete successor/transport.

4 The bridge: compiling Schrödinger's relative bifurcation into a LagEvent

4.1 Input data for compilation

From World A, take:

- $S := Ext(T')$,
- $V := Ext(T)$,
- $obs := \rho_i$,
- $s_0 := \Delta_0, s_1 := \Delta_1$,

with:

$$obs(s_0) = obs(s_1) \quad \text{and} \quad s_0 \neq s_1.$$

4.2 Universal two-state template $G(s_0, s_1)$

Construct a primitive system $E_G := G(s_0, s_1)$ with:

States: $S_G = \{s_0, s_1\}$.

Observable: $obs_G := obs|_{S_G}$.

Target:

$$target(base) := obs(s_0) \in V,$$

so $s_0, s_1 \in FiberPt(base)$ because $obs(s_0) = obs(s_1)$. Since S_G has exactly two states, the fiber contains exactly those two points.

Paths: generators $id, all, step$.

Minimal 2D data: $Deformation := True$, giving $\alpha : Deformation \text{ all } id$.

Semantics:

- $sem(id) = \text{identity relation}$,
- $sem(all) = \text{total relation}$,
- a minimal discriminating step:

$$sem(step)(x, y) :\Leftrightarrow (x = s_0 \wedge y = s_0).$$

4.3 Result: a canonical LagEvent

With existential *Compatible*:

- $Compatible(step, s_0)$ holds (witness $y = s_0$),
- $\neg Compatible(step, s_1)$ holds (no witness exists).

With the standard existential form of *HolonomyRel* (witnessed by some y) and $sem(all)$ total + $sem(id)$ identity, one obtains:

$$HolonomyRel(\alpha)(s_0)(s_1).$$

Therefore:

$$LagEvent_{E_G}(\alpha, step).$$

Define:

$$Gen(i, \Delta_0, \Delta_1, agree, nontriv) := G(\Delta_0, \Delta_1).$$

Interpretation: Schrödinger's "indistinguishable alternatives" compile into a concrete holonomic obstruction that contains a discriminating step.

5 Scaling: pushing the obstruction to richer models via **Comp**

Let E be a richer model (apparatus, observer, environment, etc.). If there exists a morphism

$$M : E_G \rightarrow E$$

in the admissible class **Comp**, then the pushforward theorem gives:

$$LagEvent_{E_G}(\alpha, step) \Rightarrow LagEvent_E(M\alpha, Mstep).$$

A minimal **Comp** specification requires:

(0) **Structural transport**:

- Sem-lax: $sem_G(p)(x, y) \Rightarrow sem_E(Mp)(Mx, My)$,
- Deformation-map: $\alpha : Deformation_G(p, q) \Rightarrow M\alpha : Deformation_E(Mp, Mq)$.

(1) **Observable compatibility** (commutation needed to map fibers),

(2) **Fiber-faithfulness** (preserve $x \neq x'$ within fibers),

(3) **Push witnesses** for positive/existential parts (Compatible, HolonomyRel witnesses),

(4) **NoNew Σ** for critical steps (push the negative: no new compatibility witnesses appear).

Existence of such an M is a modeling assumption, not automatic. The theorem states: if $M \in Comp$, then the obstruction persists at scale.

6 One-line summary

Schrödinger's cat fits the framework as:

$$(\Delta_0 \neq \Delta_1 \wedge \rho_i(\Delta_0) = \rho_i(\Delta_1)) \xRightarrow{Gen} LagEvent \xRightarrow{Comp} LagEvent \text{ in larger systems.}$$